

# JESLY PROSPER

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## PROFILE

Computer Science graduate (2:1, University of York) with two years' hands-on experience building and analysing real business systems: designed and ran the financial reporting, data infrastructure, and full-stack platform for a commercial operation end to end. Strong Python, SQL, and JavaScript, applied machine learning across classification, clustering, and deep learning, and a track record of translating technical work into decisions stakeholders acted on. Certified in business analysis, financial planning, cybersecurity, and Python, with everyday use of AI tools.

## TECHNICAL SKILLS

**Programming & Data:** Python (Flask, Pandas, NumPy, Scikit-learn, PyTorch), Java, JavaScript/TypeScript (React, Next.js), SQL (T-SQL, PostgreSQL, MS SQL Server)

**Machine Learning & Analytics:** Predictive modelling (Logistic Regression), clustering (K-Means), deep learning (CNNs, PyTorch), statistical analysis, ETL pipeline design

**Business & Financial Analysis:** Financial reporting, revenue and cost analysis, KPI tracking, forecasting support, requirements gathering, business process mapping

**Cloud & Engineering Practice:** AWS (Lambda, S3, EC2), Docker, Snowflake, Git, CI/CD pipelines, REST API design, automated testing, Agile delivery (Jira)

**Reporting & AI Tools:** Tableau, advanced Excel (pivot tables, complex formulas), everyday use of AI tools (Claude, GitHub Copilot) for development and analysis

**Languages:** Fluent in English and Haitian Creole

## EDUCATION

**BEng in Computer Science, 2:1 Honours**

June 2026

University of York, United Kingdom | Award mark 63. Relevant coursework: Data Science & Machine Learning, Autonomous Robotics, Software & Systems Engineering, Object Oriented Data Structures & Algorithms

## WORK EXPERIENCE

**Software Developer, D&K Car Rentals LTD**

June 2023 – September 2025

Grand Turk, Turks and Caicos Islands

- Designed, built, and solely maintained a full-stack booking, fleet, and financial reporting system (Python/Flask + SQL Server, later migrated to Next.js/Supabase) for a live commercial business.
- Analysed booking, revenue, and cost data to identify trends and inefficiencies, translating findings into recommendations that directly informed pricing and fleet decisions.
- Automated monthly financial reporting and recurring processes, replacing manual work and improving accuracy, consistency, and turnaround.
- Gathered requirements directly from non-technical stakeholders and delivered working solutions end to end, owning the system's full lifecycle from design to production support.

**Freelance Software Developer, Self-Employed**

March 2020 – Present

Grand Turk, Turks and Caicos Islands / remote

- Built automated data pipelines, reporting dashboards, and web applications for multiple client businesses, integrating APIs, databases, and flat files into reliable systems.
- Implemented data validation and error-handling frameworks ensuring the accuracy and integrity of every client deliverable.

## CERTIFICATIONS

**Introduction to Business Analysis — LinkedIn Learning (IIBA-accredited)**

Aug 2026

**Financial Planning and Analysis (FP&A) Foundations — LinkedIn Learning**

2026

**CFI Corporate Finance Foundations Professional Certificate — LinkedIn Learning**

2026

**Cybersecurity Job Simulation — Mastercard (Forge)**

Feb 2025

**The Legend of Python — Codédex**

Feb 2025

## PROJECTS

**Autonomous Robotics System (University of York)**

January 2026

- Designed and implemented an autonomous multi-robot barrel-collection system for TurtleBot3 Waffle Pi robots in a Gazebo simulation, using Python and ROS2 within a Dockerised development environment.
- Built an 8-state finite state machine (searching, approaching, positioning, picking up, delivering, offloading, clearing space, decontaminating) to coordinate patrol navigation, target selection, pickup, delivery, and decontamination behaviours.
- Used the Nav2 stack for waypoint-based patrol navigation and LiDAR-based obstacle sectoring for real-time obstacle avoidance, combined with a proportional controller for visual servoing to align and approach detected targets.
- Implemented hysteresis-based target selection to prevent oscillation between similarly-sized detections, plus a timeout and escape manoeuvre to recover automatically when the robot became stuck.

- Designed and ran five distinct test scenarios (baseline, sensor noise robustness, varied item distribution via random seed, open environment, and a full-complexity stress test combining noise and obstacles), each run for 3 hours of simulation time, to validate robustness under different conditions, achieving 90–100% detection accuracy across scenarios.

#### **NHS Data Science & ETL Pipeline (University of York)**

**May 2025**

- Designed a relational database schema from six raw NHS operational datasets (appointments, billing, insurance, prescriptions, surgery, test records), building an ER diagram and normalising the data into SQLite.
- Ran a chi-square test to identify significant drivers of appointment no-shows, built a logistic regression model to predict no-show likelihood (correctly identifying and flagging overfitting rather than reporting an inflated result), and applied K-means clustering to segment patients by behaviour.

#### **CIFAR-10 Image Classification (University of York)**

**May 2025**

- Built a CNN from scratch in PyTorch (three convolutional blocks with batch normalisation and dropout, trained with Adam and a scheduled learning rate over 20 epochs), achieving 81% test accuracy on a 40,000/10,000/10,000 train/validation/test split.

#### **Campus Tycoon – Java Simulation Game (University of York)**

**December 2024**

- Built a libGDX-based city simulation game in Java, using the Factory pattern to instantiate building types and inheritance/polymorphism across building and event class hierarchies, coordinating a 4-person Agile team with Jira and delivering on time with CI/CD integration.